

Walmart Store Performance & Inventory Planning Analysis

Prepared by: Saffan Nagani

Date: May 23, 2026

Tools used: SQL (DB Browser for SQLite), Microsoft Word

**Dataset: Walmart Weekly Sales Dataset - 6,435 records across
45 stores**

Section 1 – Business Problem

Walmart operates 45 stores across multiple locations with varying levels of demand. Without a clear understanding of which stores drive the most revenue, impact on sales during holiday season, and how factors such as unemployment levels affect consumer behaviour, inventory planning can become a difficult task. Also, Lack of strategic planning can increase the risk of stockouts at high demand stores and overstocking at low-demand centres.

Therefore, this analysis uses previous Walmart weekly sales data to identify demand patterns and provide actionable insights to effectively plan the inventory.

Section 2- Key findings

1. Massive revenue gap across stores

Top performing store (Store 20, \$301M total sales) generated 8x more revenue than the lowest performer (Store 33, \$37M). The level of variability in demand clearly demonstrates that a unified inventory strategy for all stores will fail and can result in losses.

2. Holidays boost demand by 8%

Holiday weeks average \$1.12M in weekly sales per store compared to \$1.04M during normal weeks which is an 8% hike. With only 450 holiday weeks out of 6,435 total records, holidays are rare but entirely predictable, therefore it is recommended to turn them into a planning opportunity to maximize profits.

3. Store 10 is the most holiday-sensitive location

Store 10 shows the largest holiday impact with a \$230K spike in weekly sales during holiday periods, followed by Stores 28, 35, 2, and 4. Therefore, these stores face the highest risk of stockouts if holiday inventory buffers are not planned accordingly.

4. Stores 14, 10, and 20 carry the highest demand volatility

Store 14 showed a \$2.3M difference between its best and worst performing week. High volatility stores cannot be managed with fixed inventory cycles. Therefore, store 14 and others in the list require a constant data-driven attention for smoother operations.

5. Unemployment significantly affects purchasing power

As is usually the case, stores in low unemployment areas average \$1.18M in weekly sales versus \$854K in high unemployment areas marking a 38% spread. Economic conditions at the store level are a key demand metric and should be incorporated into forecasting models for added benefits.

Section 3 – Recommendations

1. Tier your inventory allocation across stores

It is recommended that you do not distribute resources equally across all 45 stores. Instead, create a three-tiered system based on total sales performances:

Tier 1 (High Priority): Stores 20, 4, 14 - highest sales volume

Therefore, these stores require the highest priority with the maximum amount of stock levels.

Tier 2 (Medium Priority): Mid-range stores with standard replenishment

Use weekly cycles to monitor changes at these centres.

Tier 3 (Low Priority): Stores 33, 44, 5 - reduce inventory allocation

These stores are underperforming; therefore, it is recommended that further investigations are conducted to figure out root causes behind the low performance before investing further.

2. Build a pre-holiday inventory buffer at sensitive stores

Stores 10, 28, 35, 2, and 4 recorded a 10% increase in sales during the holiday period. These stores have demonstrated spikes in sales consistently and therefore it is important to increase inventory levels by a minimum of 10% a week prior to the holiday period at these stores.

3. Switch high-volatility stores to dynamic replenishment

Stores 14, 10, and 20 showed complex replenishment cycles during the analysis. On certain occasions, the sales swung by at least \$2M+. Hence, these stores require weekly demand monitoring to avoid stockouts or excessive inventory costs.

4. Factor unemployment data into demand forecasting

Incorporate local unemployment rates as variables in demand forecasting. Data suggests that stores in high unemployment areas consistently underperformed compared to areas with high employment rates. Inventory targets should reflect these economic factors to avoid overstocking at certain locations.

Section 4 – Methodology

Data Source: Walmart Weekly Sales Dataset (Kaggle)

Records Analyzed: 6,435 weekly sales records across 45 stores

Time Period: 2010 - 2012

Tool Used: SQL via DB Browser for SQLite

Analysis approach:

- Aggregated total and average weekly sales by each store to identify performance tiers
- Distributed data by holiday flag to measure demand spikes during the holidays
- Calculated sales range (max minus min) by each store to identify demand volatility
- Grouped unemployment rates into three brackets to analyze the relationship between economic conditions and purchasing behaviour

All queries were written from scratch in SQL. Dataset contained no missing values; therefore, no data cleaning and treatments were necessary.